4 **(a)** State what is meant by *simple harmonic motion*.

.....

.....

.....

..... [2]

(b) Fig. 4.1 shows a mass m of 0.80 kg attached to an unstretched helical spring and allowed to come to rest.

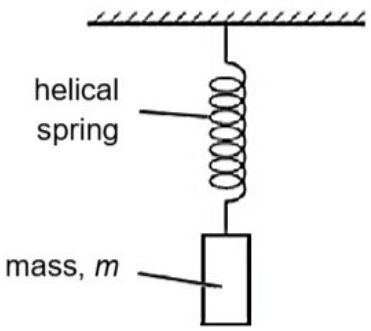


Fig. 4.1

The mass is then pulled down a distance of 1.6 cm and released, causing it to oscillate in simple harmonic motion at a frequency of 3.0 Hz.

(i) Calculate the maximum kinetic energy of the oscillations.

maximum kinetic energy = ... J [2]

(ii) Determine speed v of the mass at time t = 0.20 s.

v = ... m s^{-1} [2]